



Interactions of histidine-rich glycoprotein with immunoglobulins and proteins of the complement system

G.A. Manderson^{a,b}, M. Martin^a, P. Önnarfjörð^b, T. Saxne^c, A. Schmidtchen^d,
T.E. Mollnes^e, D. Heinegård^b, A.M. Blom^{a,*}

^a Lund University, Department of Laboratory Medicine Malmö, Division of Medical, Protein Chemistry, The Wallenberg Laboratory Floor 4, University Hospital, Malmö, Entrance 46 Malmö, SE-205 02, Sweden

^b Department of Experimental Medical Science, Division of Cell and Matrix Biology, BMC, Lund University, SE-221 84, Lund, Sweden

^c Department of Rheumatology, Lund University Hospital, SE-221 85, Lund, Sweden

^d Division of Dermatology and Venereology, BMC, Lund University, SE-221 84, Lund, Sweden

^e Institute of Immunology, University of Oslo and Rikshospitalet, Oslo, Norway

ARTICLE INFO

Article history:

Received 1 April 2009

Received in revised form 15 July 2009

Accepted 16 July 2009

Available online 11 August 2009

Keywords:

Histidine-rich glycoprotein

Immune complexes

Complement

C1q

Necrotic cells

ABSTRACT

This study describes how the serum protein histidine-rich glycoprotein (HRG) affects the complement system. We show that HRG binds strongly to several complement proteins: C1q, factor H and C4b-binding protein and that it is found complexed with these proteins in human sera and synovial fluids of rheumatoid arthritis patients. HRG also binds C8 and to a lesser extent mannose-binding lectin, C4 and C3. However, HRG alone neither activates nor inhibits complement. Both HRG and C1q bind to necrotic cells and increase their phagocytosis. We found that C1q competes weakly with HRG for binding to necrotic cells whilst HRG does not compete with C1q. Furthermore, HRG enhances complement activation on necrotic cells measured as deposition of C3b. We show that HRG inhibits the formation of immune complexes of ovalbumin/anti-ovalbumin, whilst the reverse holds for C1q. Immune complexes formed in the presence of HRG show enhanced complement activation, whilst those formed in the presence of C1q show diminished complement activation. Taken together, HRG may assist in the maintenance of normal immune function by mediating the clearance of necrotic material, inhibiting the formation of insoluble immune complexes and enhancing their ability to activate complement, resulting in faster clearance.

© 2009 Elsevier Ltd. All rights reserved.

1. Introduction

The plasma protein histidine-rich glycoprotein (HRG) is known to interact with components of the immune system. Although many studies have been published, there is so far no clear understanding of what effect these interactions have on immune function. HRG is a multi-domain 67 kDa protein (Haeberli, 1992) that is present in human plasma at concentrations between 100 and 150 µg/mL (Jones et al., 2005a). As the name suggests, the protein contains an unusually high number of solvent-exposed histidine (and also prolines residues) that are concentrated in central α-helical domains. The histidine side-chains are involved in the coordination of Zn²⁺

ions (Jones et al., 2005a). It is thought that metal binding (and also pH change) causes a conformational change that can affect ligand binding, possibly regulating the biological activity of HRG (Jones et al., 2005a).

Many of the *in vivo* ligands of HRG are potentially important for modulation of immune system function. For example, the F_{ab} region of immunoglobulin G (IgG) molecules have been shown to bind the N-terminal domain of HRG (Gorgani et al., 1999). Additional *in vitro* experiments showed that the formation of insoluble immune complexes (IICs) is partially inhibited at physiologically relevant (i.e. 150 µg/mL) concentrations of HRG (Gorgani et al., 1997).

An interaction of HRG with C1q, the initiator of the classical pathway of complement, has been documented (Gorgani et al., 1997). However, the consequences of this specific interaction on the downstream activation of complement have not been studied. Complement is part of the innate immune system and consists of three paths of proteolytic cascades that can be activated by: pathogenic microorganism surfaces (lectin pathway), immune complexes (classical pathway) and the autoactivation of the structurally unstable C3 (alternative pathway). The three pathways converge and ultimately lead to the formation of a lytic membrane attack complex (MAC). Complement activation also results

Abbreviations: HRG, histidine-rich glycoprotein; IC, immune complex; IIC, insoluble immune complex; SIC, soluble immune complex; TCC, terminal complement complex; NHS, normal human serum; FH, factor H; C4BP, C4b-binding protein; KCP, Kaposi sarcoma-associated herpes virus (KSHV) complement control protein; mAb, monoclonal antibody; pAb, polyclonal antibody; MAC, membrane attack complex; CRP, C-reactive protein; FI, factor I; BSA, bovine serum albumin; MFI, mean fluorescence intensity.

* Corresponding author. Tel.: +46 40338233; fax: +46 40337044.

E-mail address: anna.blom@med.lu.se (A.M. Blom).

in opsonisation of “self” cells and tissue such as dying cells, which become targeted for phagocytosis (Smith et al., 2002). Additionally, the products of two of the proteolysis reactions are the anaphylatoxins C3a and C5a, which trigger an inflammatory response and assist in recruiting phagocytic cells to sites of injury (Kohl, 2001). In a disease setting, proteins, such as HRG, may trigger elevated complement activation. This could ultimately lead to the establishment of a vicious circle of inflammation, cell death, the delayed clearance of cell debris, continued complement activation, and the eventual destruction of “self” tissue (Jancar and Sanchez Crespo, 2005). Additionally, when immune complexes are not cleared efficiently, they can continue to aggregate and their eventual precipitation is toxic to cells (Charlesworth et al., 1982). Only one report has investigated the link between HRG and complement in detail. It proposes a biphasic modulation by HRG of cascade activation, possibly via interactions with C8, C9, S protein and factor D (Chang et al., 1992). More work is needed to clarify the mechanistic details and consequences of the interaction of HRG with these and other complement proteins.

Also relevant to immune system function, HRG has been reported to bind to necrotic cells via the N1N2 domain and an, as yet, unidentified ligand of cytoplasmic origin (Jones et al., 2005b). As a consequence of HRG binding, enhanced phagocytosis of necrotic cells by macrophages was reported *in vitro* (Jones et al., 2005b). HRG was also found to bind to late apoptotic cells via DNA and Fcγ-receptor I on mature human monocyte-derived macrophages, which resulted in enhanced phagocytosis (Gorgani et al., 2002). Thus, through opsonisation, HRG may play a physiological role in the clearance of late apoptotic and necrotic cells by phagocytes. Interestingly, in addition to its role as the initiator of the classical pathway (Gaboriaud et al., 2004), C1q binds to apoptotic and necrotic cells (Elward et al., 2005) enhancing their rate of clearance by phagocytic cells. Little is known about the effects HRG may have on this aspect of C1q function. Additionally, because HRG, C1q and IgGs have been shown to bind to one another, a tripartite “interplay” may exist, which could modulate both complement function and immune complex (IC) growth and their complement-activating potential.

The aims of the present study are to: (1) identify HRG–complement interactions, and determine whether these interactions occur *in vivo*. (2) Continue the investigation regarding what effect HRG may have on complement function and attempt to pinpoint where, and on which pathway, HRG acts. (3) Investigate how the binding of HRG to C1q may affect the interaction of both proteins with necrotic cells, and also the formation and complement-activating ability of insoluble immune complexes.

2. Materials and methods

2.1. Materials

Samples of plasma and synovial fluid were collected from rheumatoid arthritis patients at the Department of Rheumatology, Lund University Hospital. Samples were centrifuged and stored at -80°C until analysis. All patients, $n = 20$, fulfilled the American College of Rheumatology 1987 criteria for rheumatoid arthritis (Arnett et al., 1988). Pooled normal human serum (NHS) was prepared from the freshly drawn blood of healthy volunteers. The ethical review board of Lund University has approved collection of blood and synovial fluids. C1q was purified from human serum (Tenner et al., 1981). The partial digestion of C1q using pepsin and collagenase (Worthington) was used to prepare tail (Paques et al., 1979) and head (Reid and Porter, 1976) regions. Factor H (FH), factor I (FI) and C4b-binding protein (C4BP) were purified from human plasma according to Blom et al. (2003a), Crossley and Porter (1980) and

Dahlbäck (1983), respectively. C3-met and C4-met were prepared as described in Blom et al. (2003c). Mannose-binding lectin (MBL) was purchased from Statens Serum Institut, Denmark. C-reactive protein (CRP) was from Calbiochem, IgG from Immuno and an anti-C1q monoclonal antibody (A201) was from Quidel. C8 and C9 were from Complement Technology, Inc. Chicken ovalbumin was obtained from Sigma–Aldrich. Fibromodulin was prepared as described in Heinegård et al. (1986). An anti-HRG monoclonal antibody (mAb, HRG-5, G-14-3-4D5/65) was provided by Dr C Parish, Australian National University, Australia and anti-HRG polyclonal antibody (pAb, GHH 20 3) by Dr A Schmidtchen, Biomedical Centre, Lund University, Sweden. Polyclonal anti-factor H (FH) and anti-chicken ovalbumin antibodies were from AbCam. Anti-Kaposi sarcoma-associated herpes virus (KSHV) complement control protein (KCP) and C4BP (PK 8604) pAbs were prepared in-house. The antibodies aE1195B and 9C4-RQ-B were from Dr TE Mollnes, The National Hospital, Oslo, Norway. All other antibodies, and a horseradish peroxidase streptavidin conjugate were from Dako. Fresh sheep erythrocytes were purchased from Swedish Veterinary Institute and Jurkat T-cells were from the American Type Culture Collection (ATCC).

2.2. Purification of HRG

Human HRG was purified from defatted plasma by Ni-NTA (Qia-gen) affinity chromatography. Plasma was loaded onto a column pre-equilibrated in TBS, pH 8.0, 1 mM EDTA, 100 $\mu\text{g}/\text{mL}$ PMSF and 2 $\mu\text{g}/\text{mL}$ aprotinin. Weakly bound material was eluted using the same buffer with 150 mM imidazole and HRG was eluted using the first buffer with 500 mM imidazole. Pure fractions were dialysed against TBS, pH 7.4, concentrated, by stirred cell ultrafiltration to approximately 1 mg/mL, then frozen at -80°C .

2.3. Quantitation of concentrations of HRG in serum and synovial fluid

Determinations were made by ELISA. A standard curve was prepared using a serum pool from healthy volunteers. Microtitre plate (Nunc Maxisorp) wells were coated overnight at 4°C , using 50 μL of anti-HRG (αHRG) pAb, diluted with coating buffer (75 mM Na_2CO_3 , pH 9.6), then washed 3 times (TBS, 2 mM CaCl_2 , 0.1% Tween 20), as was done after all incubation steps. Plates were then blocked (200 μL of 1% BSA in washing solution) at room temperature for 2 h. Subsequent incubations were, unless stated otherwise, made at room temperature for 1 h using 50 μL volumes. Plates were incubated with parallel plasma and synovial fluid samples, each from 20 rheumatoid arthritis patients, diluted in binding buffer (50 mM HEPES, pH 7.4, 100 mM NaCl, 2 mM CaCl_2). Next, plates were incubated with a αHRG mAb, and then a goat- α -mouse horseradish peroxidase (HRP) conjugate (30 min incubation), both diluted in blocking solution. Bound HRG was detected using 1,2-phenylenediamine dihydrochloride (OPD)/ H_2O_2 (Dako). The reaction was stopped using 0.5 M H_2SO_4 and A_{490} values determined.

2.4. Binding of HRG to complement-related ligands

Details as per the above ELISA protocol. Microtitre plate wells were coated overnight with: C1q, FH, C4-met, C3-met, FI, C4BP, IgG, CRP, C8, C9, MBL, fibromodulin and BSA at 6 $\mu\text{g}/\text{mL}$. Following blocking, the plates were incubated with HRG diluted with binding buffer to 1–20 $\mu\text{g}/\text{mL}$. Bound HRG was detected as above. The effect of NaCl on the binding of HRG to C1q and proteolytically cleaved C1q heads and tails was also examined. C1q species were coated overnight at 10 $\mu\text{g}/\text{mL}$. Plates were then incubated with 10 $\mu\text{g}/\text{mL}$ HRG diluted using various ratios of binding buffer and 4 M NaCl to achieve NaCl concentrations between 100 and 850 mM.

2.5. Identification of HRG in complex with other proteins in sera

An immune precipitation approach was used. Initially, serum or synovial fluid (100 μ L aliquots) was pre-cleared by incubating with an irrelevant α FI pAb (20 μ g) at 4 °C (the incubation temperature for all subsequent steps) for 60 min. Any protein/Ig complexes formed were removed by adding 4 μ L protein A Sepharose resin (Sigma) mixed with 16 μ L Sephacryl S-100 resin (GE Healthcare) and incubating, with rotating, for 60 min. Subsequently, 20 μ g of the following pAbs were added to 100 μ L aliquots of fresh sera: α HRG, α C1q, α FH, α C4BP, α KCP, followed by overnight incubation, with rocking. Fresh protein A Sepharose (as above) was then added followed by incubation, with rotating, for 60 min. The resin was sedimented, retained, washed 3 times in 400 μ L cold microtitre plate washing solution (see above), and re-suspended in 45 μ L SDS-PAGE sample buffer. Precipitated material was analysed by Western blot using a α HRG mAb, then a goat- α -mouse pAb HRP conjugate. Protein bands were visualized using a luminol enhanced chemiluminescence substrate system (Sigma–Aldrich, St. Louis, Missouri).

The binding of HRG from serum to C1q, FH, IgG and C8 was also investigated. HRG ligands were immobilised at 10 μ g/mL, then incubated at 37 °C for 60 min with NHS, diluted 1:10 in binding buffer. Bound HRG was detected using the α HRG pAb.

2.6. Direct activation of complement by immobilised HRG

Microtitre plate wells were coated overnight with HRG, IgG, or BSA at 5 μ g/mL. Following blocking, plates were incubated with normal human serum (NHS), diluted in GVB++ (5 mM veronal buffer, pH 7.35, 144 mM NaCl, 0.1% gelatin, 1 mM $MgCl_2$, 0.15 mM $CaCl_2$) as follows: deposition of C3d, 1:200 and C5b–C9, 1:10. Serum incubations were done at 37 °C for 20 min (C3d) or 40 min (C5b–C9). For detection, plates were incubated with pAbs against C3d and C5b–9, then a swine- α -rabbit pAb HRP conjugate.

2.7. Inhibition of complement activation by HRG

Microtitre plate wells were coated overnight with fibromodulin at 10 μ g/mL, then blocked. NHS was mixed with HRG and GVB++ to achieve a final serum dilution of 1:50 and HRG between 2.5 and 25 μ g/mL. The volume of HRG solution, relative to GVB++ and serum never exceeded 7%. These serum/HRG mixtures were incubated at 4 °C for 1 h before being used in complement deposition assays (see above).

2.8. Effect of HRG on complement-induced erythrocyte haemolysis

Sheep erythrocytes were washed 3 times in DGVB++ (2.5 mM veronal buffer, pH 7.35, 72 mM NaCl, 0.1% gelatin, 2.5% D-glucose, 1 mM $MgCl_2$, 0.15 mM $CaCl_2$), then adjusted to 1×10^9 cells/mL. Next, the cells were mixed with an equivalent volume of amboceptor (Dade Behring), previously diluted 1:1500 in DGVB++, and incubated at 37 °C for 20 min. The erythrocytes were washed twice, then re-suspended to 5×10^8 cells/mL. For haemolytic assays, erythrocytes were diluted to 8.6×10^7 cells/mL in DGVB++, then effectors added: HRG (to 0.340 μ M), C1q (to 0.056 μ M), FH (to 0.103 μ M) and C4BP (to 0.038 μ M). When two proteins were added simultaneously, they were pre-incubated at 4 °C for 5 min. Haemolysis was initiated by adding NHS, to a final dilution of 1:270 and the process followed, at 37 °C, by measuring the decrease in light scattering (A_{700}). For HRG alone, the protein was also pre-incubated with NHS (37 °C, 10 min) before being added to the erythrocytes.

Pre-mixed HRG and C1q were also added to C1q-depleted serum, which was then used for haemolytic assays. HRG (554 pmol) and C1q (45 pmol) were pre-incubated for 30 min at 4 °C, then added to

C1q-depleted serum to give 80 μ g/mL C1q and 125 μ g/mL additional HRG. The depleted serum was then diluted 1:80, and 25 μ L was added to 125 μ L sheep erythrocytes at 8.6×10^7 cells/mL in DGVB++. Extent of haemolysis was assessed spectrophotometrically from amounts of liberated haemoglobin (A_{405}).

2.9. Competition between HRG and C1q for binding to necrotic cells

A flow cytometry approach was employed. Jurkat T-cells (ATCC) in RPMI without serum at 2×10^6 cells/mL were heat-treated at 56 °C for 30 min to induce necrosis. Cells were then washed in flow cytometry binding buffer (10 mM HEPES, pH 7.4, 150 mM NaCl, 5 mM KCl, 1 mM $MgCl_2$, 2 mM $CaCl_2$) and re-suspended to 1.5×10^7 cells/mL. For titration of HRG and C1q with necrotic cells, protein and cells were incubated together at 37 °C for 30 min with gentle shaking. Following washing in flow cytometry binding buffer, the cells were incubated (30 min, room temperature) with a 1:1000 dilution of a α HRG mAb or a α C1q pAb, washed again, then incubated with a 1:40 dilution of a goat- α -mouse or swine- α -rabbit pAb FITC conjugate. The cells were washed and re-suspended in flow cytometry analysis buffer (50 mM HEPES, pH 7.4, 100 mM NaCl, 30 mM NaN_3 , 1% BSA) and the quantities of bound HRG or C1q determined.

For competition studies, C1q labelled with AF488 (Molecular Probes) according to the manufacturer's instructions, and unlabelled HRG, or alternatively unlabelled C1q and AF488-HRG were mixed and pre-incubated at 4 °C for 30 min. Necrotic Jurkat T-cells were added and incubated with the proteins at 37 °C for 30 min with gentle shaking. Following washing with flow cytometry binding buffer, the cells were re-suspended in flow cytometry analysis buffer and the quantities of the bound labelled proteins determined. The mole ratios AF488-C1q:HRG of 1:0, 1:1, 1:2 and 1:4, correspond to 7 μ g/mL C1q: 0, 1, 2.4, 4.2 μ g/mL HRG. For AF488-HRG:C1q, the mole ratios 1:0, 1:0.33, 1:0.66, 1:1 and 1:2, correspond to 15 μ g/mL HRG: 0, 15 33.3, 66.6, 100 and 200 μ g/mL C1q.

2.10. Confocal microscopy

Necrosis was induced in RPMI without FCS as above. After washing the cells twice with binding buffer (10 mM HEPES, 150 mM NaCl, 5 mM KCl, 1 mM $MgCl_2$, 2 mM $CaCl_2$), 1×10^6 cells were incubated with a combination of 350 μ g/mL C1q and 50 μ g/mL HRG (equimolar amounts) for 30 min at 37 °C. The cells were transferred to ice and washed twice with ice-cold binding buffer and spun for 5 min at 500 rpm on SuperFrostPlus object slides (Menzel, Germany) using a Cellspin (Tharmac, Germany). After fixation with 3.7% PFA for 15 min at room temperature, the cells were blocked for 30 min with 1% BSA in PBS. The cells were next incubated with rabbit α -human HRG and mouse α -human C1q over-night at 4 °C, then with AF488-labelled goat anti-rabbit and AF647-labelled goat anti-mouse, as appropriate, for 1 h at room temperature. Finally, 100 ng/mL propidium iodide (PI) was added and the cells incubated for 15 min at room temperature. In between each stop, the slides were washed extensively with PBS, then dipped briefly in 99% ethanol. After complete drying, the samples were mounted with DakoCytomation Fluorescent Mounting Medium (Dako) and sealed with nail polish. The slides were stored at 4 °C.

2.11. Complement activation by necrotic cells in the presence of HRG

Necrotic Jurkat T-cells were prepared as described above, re-suspended to 1.5×10^7 cells/mL, then incubated with an equivalent volume of 250 μ g/mL HRG in binding buffer. After washing, as above, the cells were incubated with 0.25–5.0% NHS in GVB++

at 37 °C for 30 min. The cells were washed again and incubated (30 min, room temperature) with a 1:200 dilution of a α C3c-FITC pAb, washed again, re-suspended in flow cytometry analysis buffer and the quantities of deposited C3b determined using flow cytometry.

2.12. Effect of HRG and C1q on the formation of insoluble immune complexes

The rate of growth of ovalbumin/ α -ovalbumin IICs was followed, spectrophotometrically, in the presence or absence of HRG and C1q as described (Easterbrook-Smith et al., 1988). All time-course measurements were made in a quartz cuvette at 37 °C in filtered PBS, pH 7.4. Anti-ovalbumin was diluted to 200 μ g/mL in the presence or absence of 125 μ g/mL HRG, 15 or 30 μ g/mL C1q and the systems left to equilibrate for 5 min. Reactions were initiated by adding freshly dissolved ovalbumin, to 10 μ g/mL and the increase in light scatter was monitored (A_{350}). The final reaction volume was 80 μ L and the molar ratio ovalbumin: α -ovalbumin of 1:6 worked best.

2.13. Activation of complement by insoluble immune complexes formed in the presence of HRG and C1q

The abilities of IICs formed in the presence or absence of HRG or C1q to activate complement were assessed by measuring concentrations of the Terminal Complement Complex (TCC) using ELISA (Mollnes et al., 1985). IICs were prepared fresh, as above, by mixing 200 μ g/mL α -ovalbumin and 10 μ g/mL ovalbumin with 0, 125, 250 and 500 μ g/mL HRG, or 0, 15, 30 and 60 μ g/mL C1q, or both 250 μ g/mL HRG and 30 μ g/mL C1q, or both 125 μ g/mL HRG and 100 μ g/mL C1q. The IICs were then diluted 1/25 in GVB++ containing NHS to a final dilution of 1/25. These samples were incubated at 37 °C for 40 min. To assay for formation of the TCC, the incubations were diluted 1/25 with AG buffer (PBS, 0.02% NaN_3 , 0.2% Tween 20, 20 mM EDTA) and 50 μ L dispensed into microtitre plate wells and incubated on ice for 60 min. Microtitre plate wells had been previously coated overnight at room temperature with a 1/10 000 dilution of the Ab aE1195B (Mollnes, 1985) in PBS/0.02% NaN_3 . Following washing with PBS, 0.2% Tween 20 (PBST, used after all incubation steps), wells were incubated with 50 μ L of a 1/5000 dilution (in PBST) of the biotinylated Ab 9C4-RQ-B (Mollnes, 1985) at 37 °C for 45 min, then with 100 μ L of a streptavidin/HRP conjugate (Dako) at 37 °C for 60 min. Bound TCC was detected using the Dako OPD colourimetric assay system. A standard curve of zymosan activated serum (where the TCC concentration was defined as 1000 arbitrary units per mL, AU/mL) was used (Mollnes, 1985). The final TCC quantities in samples were expressed as AU/mL.

3. Results

3.1. Determination of HRG concentrations in the synovial fluid of rheumatoid arthritis patients

A possible link between HRG and complement was identified following the observation that HRG was one of the most abundant proteins purified from the synovial fluid of rheumatoid arthritis patients by affinity chromatography using C1q immobilised on Affi-gel (BioRad) (data not shown). HRG concentrations in both the plasma and synovial fluid of these patients were determined by ELISA. Fig. 1 shows that the concentration of HRG in plasma (199 ± 67 μ g/mL) was significantly higher than that in synovial fluid (120 ± 62 μ g/mL). Furthermore, there was only a moderate correlation between these two values for each patient ($\rho = 0.4466$, $p = 0.0484$). This would be consistent with the filtration of HRG into synovial fluid from plasma, but does not rule out that there is also some local production in the joint.

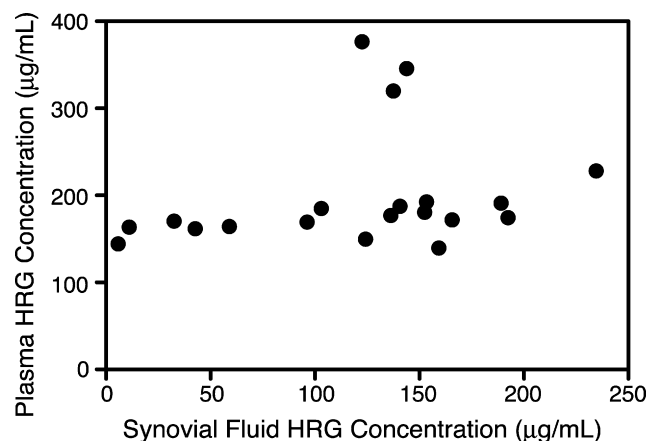


Fig. 1. Correlation of concentrations of HRG in plasma and synovial fluid from rheumatoid arthritis patients, determined by ELISA. Non-parametric Spearman correlation: $\rho = 0.4466$, with $p = 0.0484$. Mean HRG concentrations: plasma, 199 ± 67 (μ g/mL), synovial fluid, 120 ± 62 (μ g/mL), $p = 0.0001$, Wilcoxon matched-pairs test. All statistical calculations performed using InStat 3 (GraphPad).

3.2. Binding of HRG to complement-related ligands

Fig. 2A shows that HRG was successfully purified to homogeneity from human plasma and its identity confirmed by detection with a specific antibody. Coomassie staining showed a trace of aggregated material in the preparation. No attempt was made to remove this material, given its lower abundance relative to the faster-moving HRG band. Material purified as shown was used for all subsequent studies.

The interaction between HRG and purified complement proteins was examined. Immobilised complement proteins were incubated with soluble HRG so that all binding interactions could be assessed using the same α HRG mAb. HRG showed selective and strong binding to IgG and the following complement proteins: C1q, FH, C8 and C4BP (Fig. 2B and C). For 6 μ g/mL immobilised ligand (assuming complete immobilisation) and 20 μ g/mL soluble HRG (assuming complete binding), this represents molar ratios of C1q:HRG (1:24), FH:HRG (1:8), C8:HRG (1:8), IgG:HRG (1:8) and C4BP:HRG (1:29). Binding to C4, C3 and MBL was also observed. No binding of HRG to FI, CRP, C9, BSA or fibromodulin was seen. With the exception of C8, all of the curves in Fig. 2 reach a plateau, suggesting that maximal binding of HRG to the corresponding ligands had been achieved at 10 μ g/mL HRG.

The binding of HRG to isolated head and tail regions of C1q was also examined. Fig. 2D demonstrates that HRG bound mainly to the tail region and that this apparent affinity was essentially the same as that for the intact C1q molecule. The interaction between C1q and HRG also exhibited a moderate dependence on salt concentration. This suggested that ionic as well as hydrophobic interactions were involved in maintaining the interaction between HRG and C1q.

3.3. Identification of HRG in complex with other proteins in sera

Having established that HRG binds strongly and selectively to some complement proteins, we investigated whether HRG is complexed with C1q, FH and C4BP *in vivo*. Sera from healthy controls and synovial fluid from rheumatoid arthritis patients were incubated with pAbs against HRG (positive control), C1q, FH, C4BP and also KCP (negative control). Immunoprecipitation of these ligands followed by Western blot for HRG revealed that C1q, FH and C4BP were found in complex with HRG in both NHS and the synovial fluid (Fig. 3). No HRG was detected in control samples treated with TBS or immunoprecipitated using an irrelevant control pAb to KCP, demonstrating specific interactions with the above ligands. Addi-

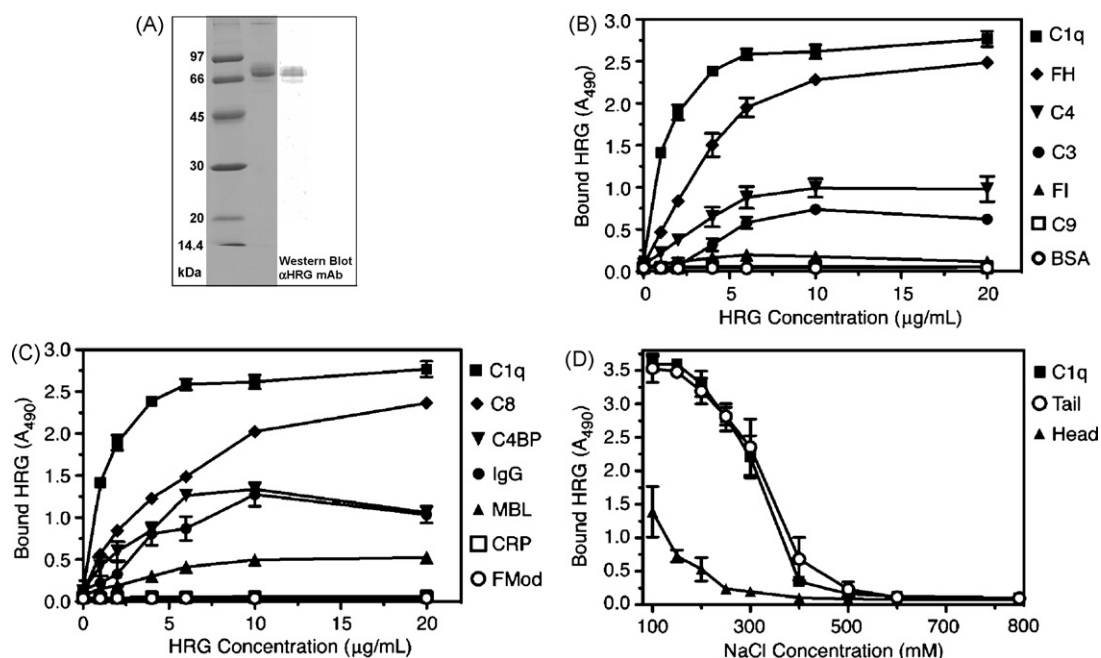


Fig. 2. Binding of HRG to complement proteins. (A) SDS-PAGE and Western Blot (using a α HRG mAb) of purified HRG. (B and C) HRG binding to complement proteins. The complement proteins indicated were coated in microtitre plate wells and incubated with increasing amounts of HRG. Bound HRG was detected using a α HRG mAb. The same C1q data are shown in both B and C as a reference ligand. (D) Dependence of the HRG–C1q interaction on NaCl. Full-length C1q and isolated heads and tails were immobilised and then incubated with HRG diluted to 10 μ g/mL with a mixture of binding buffer and 4M NaCl to achieve the concentrations of NaCl indicated. Bound HRG was detected as above. The data were collected from two representative duplicate experiments ($n=4$). Error bars denote standard deviation.

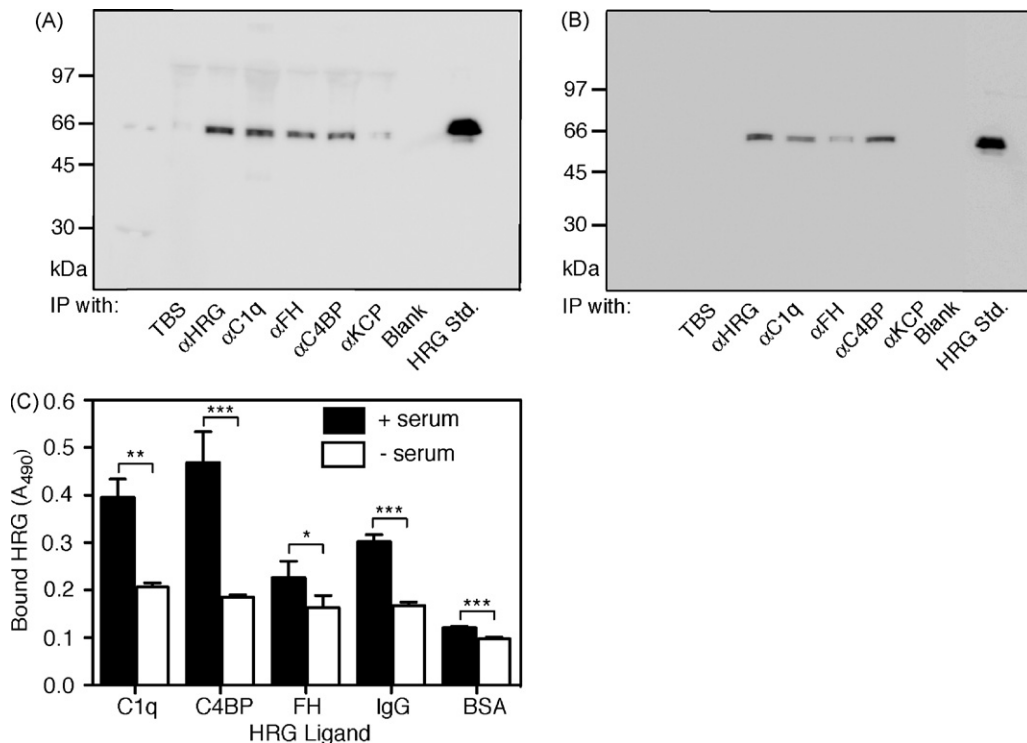


Fig. 3. Detection of HRG in complex with complement proteins in sera and synovial fluids. Immune precipitation (A and B). Aliquots of serum (A, 100 μ L) or synovial fluid (B, 100 μ L) were mixed with 20 μ g of pAbs against: C1q, FH, C4BP, HRG (positive control), KCP (negative control) or TBS, then incubated overnight. Protein A Sepharose (20 μ L) was then added to each sample, which were incubated for 60 min. The protein A Sepharose resin was retained, washed and mixed with 45 μ L SDS-PAGE sample buffer. Immune precipitated HRG was analysed by Western blot using a α HRG mAb. ECL image visualisation. (C) Binding of HRG from serum to immobilised C1q, C4BP, FH and IgG. HRG ligands were coated at 10 μ g/mL, then incubated at 37 $^{\circ}$ C for 60 min with NHS, diluted 1:10 in binding buffer. Bound HRG was detected using the α HRG pAb. The data were collected from two representative duplicate experiments ($n=4$). Error bars denote standard deviation. *, ** and *** denote statistically significant differences, $p < 0.05$, 0.01 and 0.001, respectively, relative to background Ab binding, using Student's t test.

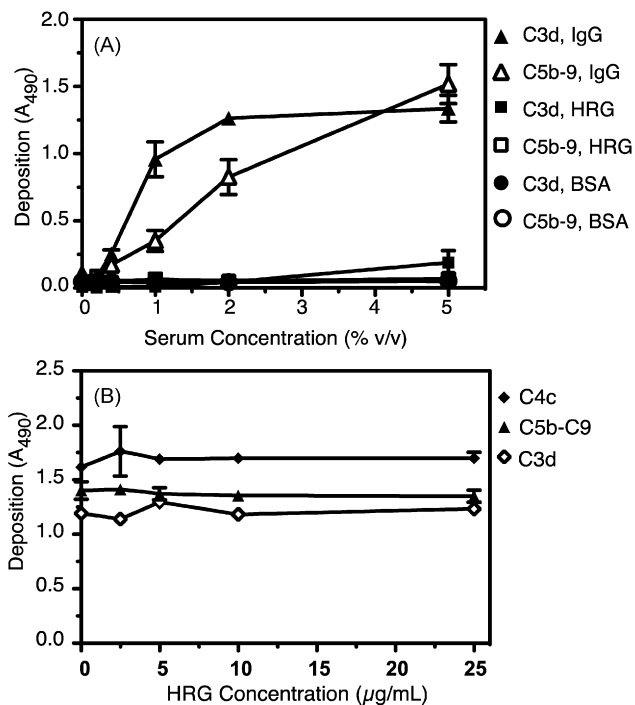


Fig. 4. Activation and inhibition of complement by HRG. (A) Activation. HRG, IgG (positive control) and BSA (negative control) were coated in microtitre plate wells. NHS was diluted with GVB++, as indicated, then incubated with the immobilised proteins for 20 min (C3d deposition), or 40 min (MAC deposition). Complement deposition products were detected with a α C3d pAb and a C5b–C9 mAb. (B) Inhibition. Microtitre plate wells were coated with fibromodulin. NHS was diluted with GVB++ as follows: C4c detection – 1:500, C3d – 1:200, C5b–C9 – 1:10, pre-mixed with HRG to the concentrations indicated and pre-incubated at 4 °C for 1 h, then 20 min (or 40 min, for C5b–C9) at 37 °C. Complement deposition products were detected as above. The data were collected from two representative duplicate experiments ($n = 4$). Error bars denote standard deviation.

tionally, from a comparison of pre-cleared and non-pre-cleared sera, no appreciable background IgG binding was apparent (data not shown).

In a related investigation, immobilised C1q, FH, C4BP and IgG were incubated with NHS and the binding of HRG was examined. All four immobilised ligands are shown to bind serum HRG (Fig. 3C), consistent with the occurrence of these interactions *in vivo*.

3.4. Direct activation and inhibition of complement by HRG

Given that HRG was found to bind to complement proteins *in vivo*, the possibility that these interactions modulate complement function was investigated. First, the activation of complement by HRG was studied. Immobilised HRG was incubated with NHS and the deposition of mid (C3d) and late (C5b–9) complement cascade reaction products were assessed. In contrast to activation and substantial complement product deposition by aggregated human IgG, no deposition, and thus complement activation, was seen for HRG (Fig. 4A). Thus, HRG alone appeared incapable of activating any of the three pathways of complement.

Next, we investigated whether HRG could inhibit the fibromodulin-induced activation of the classical and alternative pathways of complement (Sjöberg et al., 2005). Fibromodulin was used here because it does not bind to HRG, in contrast to IgG (Fig. 2). If aggregated IgG had been used to activate complement, data would reflect binding phenomena rather than deposition of complement activation products. Fig. 3B indicates that HRG, up to 25 μg/mL, did not inhibit the deposition of C4b, C3b or C5b–9. At this concentration, the binding of HRG to most complement proteins should have attained saturation (see Fig. 2). Unchanged

quantities of deposited C3b and C5b–9 suggested that HRG exerted no inhibitory effect on the classical and alternative pathways. HRG also failed to affect quantities of C3b deposited as a consequence of lectin pathway activation by mannan (data not shown). Thus, it appears that HRG exerted no inhibitory effect on any of the complement pathways.

3.5. Effect of HRG on complement-induced erythrocyte haemolysis

The effect of HRG on the ability of NHS to lyse sheep erythrocytes was examined. These investigations were conducted to corroborate the above results. Fig. 5A indicates that raising the HRG concentration in the haemolytic assay reaction volume by 0.34 μM had no effect on the rate of complement-induced haemolysis. This was true when HRG and serum were added directly to the erythrocytes without any pre-incubation. Additionally, no effect was seen when the HRG had been pre-incubated with the serum for 10 min at 37 °C before being added to the cells. Fig. 5A therefore demonstrates that HRG alone exerted no modulatory effect on the rate of complement-induced erythrocyte haemolysis.

We next explored whether the binding of HRG to C1q, FH and C4BP affected complement-induced erythrocyte haemolysis. Adding 0.056 μM C1q to the assay reaction volume attenuated the rate of erythrocyte haemolysis (Fig. 5B). This suggested that the added C1q had bound to the opsonised erythrocytes, reducing the quantity of IgG available to bind active C1, which could in turn activate complement. When 0.056 μM C1q and 0.34 μM HRG were added together, the observed rate of haemolysis was equivalent to that seen for added C1q alone (Fig. 5B). This suggested that the presence of HRG did not affect the ligand recognition functions of C1q. For addition of the complement inhibitors FH (an increase of 0.103 μM) and C4BP (an increase of 0.038 μM) slower rates of haemolysis were observed (Fig. 5C and D, respectively). When 0.103 μM FH together with 0.34 μM HRG and 0.038 μM C4BP together with 0.34 μM HRG were added, haemolysis rates were similar to those seen for the addition of only FH or only C4BP.

The effects of the interaction between HRG and C1q were examined further by pre-mixing these two proteins, adding them at physiological concentrations to C1q-depleted serum and using this serum, and a parallel C1q-repleted serum, in haemolytic assays. Rates and extents of haemolysis were the same, irrespective of whether the C1q used to replete the serum had been pre-incubated with HRG (Fig. 5E). Thus, although HRG bound strongly to C1q, FH and C4BP (Fig. 2), these interactions had a negligible effect on the function of these complement proteins.

3.6. Competition between HRG and C1q for binding to necrotic cells

Both HRG (Jones et al., 2005b) and C1q (Elward et al., 2005) bind to necrotic cells. Therefore, we investigated whether the binding of one of these proteins affected the binding of the other. Both HRG and C1q were seen to bind to necrotic Jurkat T-cells in a dose-dependent manner (Fig. 6A and B). For competition studies, AF488-C1q and AF488-HRG were used. Both of these proteins bind IgGs (Gaboriaud et al., 2004; Gorgani et al., 1999) thus complicating the interpretation of flow cytometry data. Fig. 5C demonstrates that unlabelled HRG did not compete with the labelled C1q for binding, even at a 4-fold molar excess over the C1q. Fig. 6D shows that AF488-HRG was bound to necrotic Jurkat T-cells and weak competition was seen with unlabelled C1q. The difference in mean fluorescence intensity (MFI) values between the sample without C1q and those samples with C1q was significant, although the progressive shift in MFI with

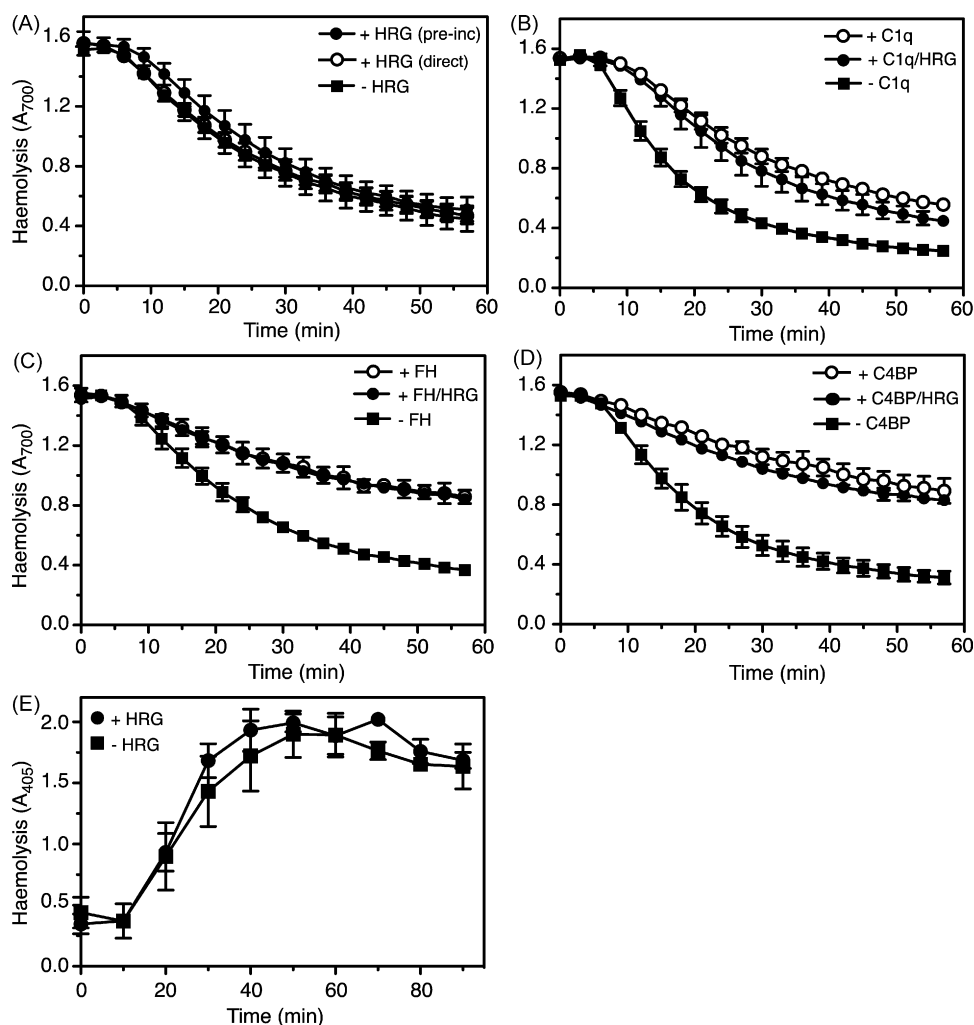


Fig. 5. Effect of HRG on complement-induced erythrocyte haemolysis. Sheep erythrocytes were opsonised with amboceptor and incubated at 37 °C: (A) with and without additional HRG (0.34 μ M), where the HRG and serum were added directly to the cells, or where the HRG and serum were pre-incubated at 37 °C for 10 min before being added to the cells, (B) with and without additional C1q (0.056 μ M) and HRG (0.34 μ M), (C) with and without additional FH (0.103 μ M) and HRG (0.34 μ M) and (D) with and without additional C4BP (0.038 μ M) and HRG (0.34 μ M). Haemolysis was assessed by measuring the decrease in light scattering at 700 nm with time. The data were averaged from 4 separate time-course experiments. Error bars denote standard deviation. (E) Pre-mixed HRG (or buffer) and C1q were added to C1q-depleted serum, to 80 μ g/mL C1q and 125 μ g/mL additional HRG, and used in haemolytic assays, where the extent of lysis was assessed from the quantity of liberated haemoglobin (A_{405}).

increasing C1q concentration was not. Thus, because C1q and HRG compete, at best, very weakly with each other for binding to necrotic cells, it is most likely that they bind to different cell surface ligands. Additionally, HRG binding was not enhanced in the presence of C1q, and *visa versa* despite the fact that these proteins interact with each other. The confocal microscopy images in Fig. 6E–I confirm that both HRG and C1q were able to bind simultaneously to necrotic cells.

3.7. Complement activation by necrotic cells in the presence of HRG

Complement is activated by necrotic cells (Ciurana et al., 2004) and the ability of cell-bound HRG to modulate this activation was examined by measuring quantities of deposited C3b. Fig. 6J shows that C3b deposition increased with increasing NHS incubation concentration. Additionally, for necrotic cells pre-treated with HRG, a statistically significant enhancement in C3b deposition was seen following incubation with 0.25% and 1% NHS. This enhancement was not seen at the higher NHS concentrations of 2 and 5%.

3.8. Effect of HRG and C1q on the formation of insoluble immune complexes

As reported earlier by Gorgani et al. (1997) and Easterbrook-Smith et al. (1988), the formation of IICs of ovalbumin and α -ovalbumin can be followed over a period of minutes as an increase in light scattering at 350 nm (Fig. 7A). When HRG was added to the system, to the physiological concentration of 125 μ g/mL, the rate of IIC formation was diminished. Additionally, the final A_{350} values attained were significantly lower. This suggests that ovalbumin/ α -ovalbumin immune complexes formed in the presence of HRG had not attained the size of those formed in the absence of HRG. The reverse was true for C1q. Significantly higher A_{350} values were reached at a faster rate. Furthermore, raising the C1q concentration, from 15 to 30 μ g/mL, resulted in an approximate doubling of the final A_{350} value. Thus, in agreement with the earlier reports (Easterbrook-Smith et al., 1988; Gorgani et al., 1997), C1q appeared to enhance rates and extents of ovalbumin/ α -ovalbumin IIC formation, whereas HRG exhibited an inhibitory effect.

For IICs formed in the presence of both HRG and C1q, data were similar to those obtained for added C1q alone. Thus, HRG was unable to perturb IIC formation when C1q was present.

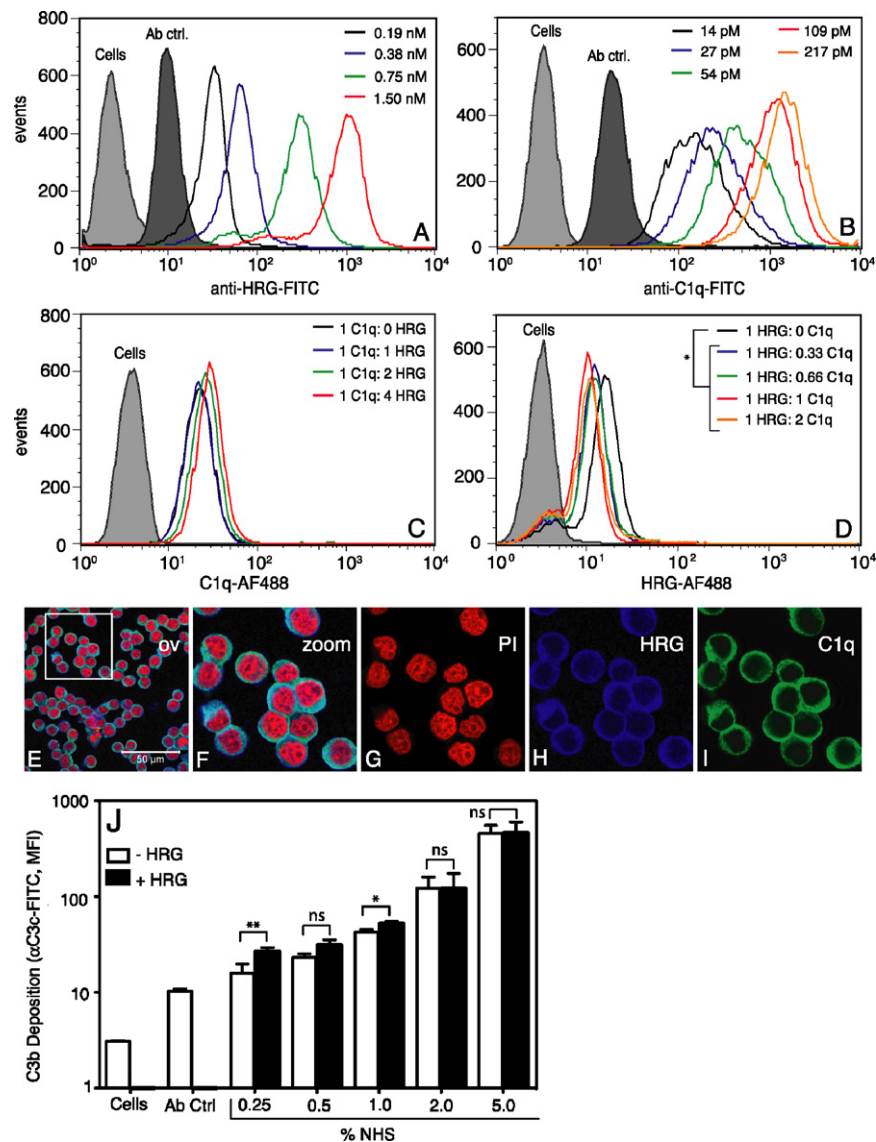


Fig. 6. Binding of HRG and C1q to necrotic Jurkat T-cells. HRG (A) or C1q (B) were mixed with necrotic cells, then incubated at 37 °C for 30 min with gentle shaking. Following washing, the cells were incubated at room temperature with a α HRG mAb or α C1q mAb, then with a goat- α -mouse pAb or swine- α -rabbit FITC conjugates and quantities of bound protein determined by flow cytometry. Competition between (C) AF488-C1q, and unlabelled HRG and (D) unlabelled C1q and AF488-HRG for binding to necrotic cells. The proteins were pre-mixed at the mole ratios indicated, pre-incubated for 30 min at 4 °C, then necrotic Jurkat T-cells were added, followed by incubation at 37 °C for 30 min with gentle shaking followed by flow cytometry analysis. * denotes statistically significant differences, $p < 0.05$, between the C1q-containing samples relative to that where C1q was absent, using Student's t test. (E–I) Confocal microscopy images of necrotic Jurkat T-cells stained with propidium iodide, α -HRG, and α -C1q (E and F), or with only propidium iodide (G), α -HRG (H), or α -C1q (I). (J) Effect of HRG on complement activation by necrotic Jurkat T-cells. HRG (125 μ g/mL) and cells were pre-mixed, washed, then incubated with 0.25–5.0% at 37 °C for 30 min, washed again and cell-bound C3b measured by flow cytometry. The data were collected from two representative duplicate experiments ($n = 4$). Error bars denote standard deviation. * and ** denote statistically significant differences, $p < 0.05$ and < 0.01 , respectively, relative to complement activation without HRG pre-incubation, using Student's t test.

When harvested by centrifugation and washed, IICs formed in the presence of HRG and C1q bound α HRG mAb (ELISA A_{490} of 2.95 ± 0.1) and α C1q pAb (ELISA A_{490} of 2.08 ± 0.1), respectively. In contrast, IICs formed in the absence of HRG and C1q gave ELISA A_{490} values of no more than 0.1 ± 0.02 with these antibodies. Thus, HRG and C1q are incorporated into, or reside on the surface of the IIC matrix.

3.9. Activation of complement by insoluble immune complexes formed in the presence of HRG and C1q

We next examined whether preparing IICs in the presence or absence of C1q and HRG would affect their ability to activate complement. Activation was assessed by measuring quantities of

the TCCs formed by incubating pre-formed IICs with NHS. Initially, controls where pre-existing levels of TCC in the NHS samples to be used were determined. Values for A_{490} , for non-activated NHS were not greater than those for the reagent blank (data not shown). This suggested that, prior to any complement activation, the NHS used in this study contained negligible quantities of TCC. Additional controls, where individual assay components, including α -ovalbumin, were pre-incubated with NHS, showed A_{490} values similar to those of the assay reagent blank (data not shown). Thus, none of the individual proteins used to prepare IICs activated complement to any appreciable extent. In contrast, when α -ovalbumin and ovalbumin were pre-incubated together before being incubated with NHS, a statistically significant increase in the concentration of TCC was seen (Fig. 7B). This is consistent

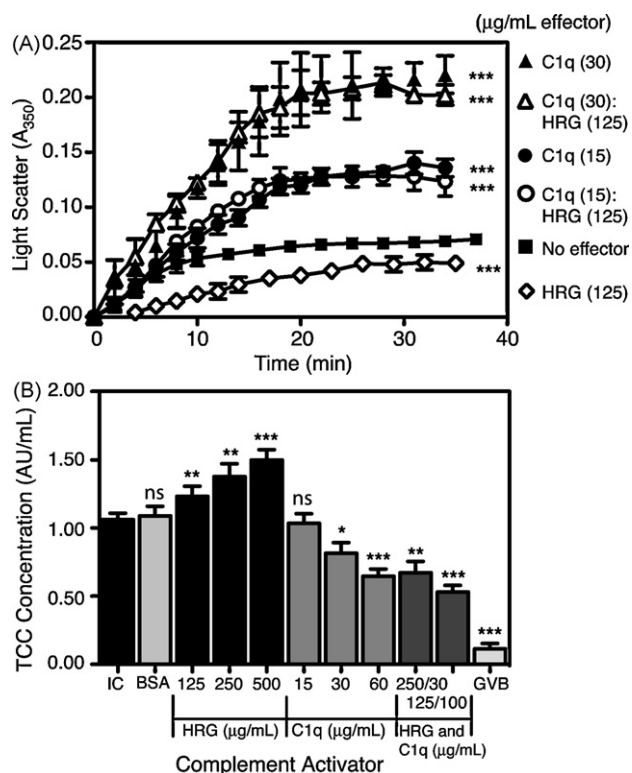


Fig. 7. Immune complexes and complement activation. (A) Formation of ovalbumin/ α -ovalbumin immune complexes in the presence or absence of HRG and C1q. α -Ovalbumin was diluted, in PBS, pH 7.4, to 200 μ g/mL with or without HRG and/or C1q, as indicated, and the IC formation reaction was initiated by adding freshly prepared ovalbumin, to 10 μ g/mL. The increase in particle size was monitored by following increases in A_{350} . The data were averaged from at least 3 separate time-course experiments. Error bars denote standard deviation. *** denotes statistically significant differences between final A_{350} values, relative to IICs prepared in the absence of HRG or C1q, $p < 0.001$, using Student's t test. (B) Activation of complement by immune complexes formed in the presence or absence of HRG and C1q. Immune complexes, prepared as above with BSA (negative control) HRG, C1q or both (as indicated) were incubated with NHS and extents of complement activation (assessed as TCC formation) were measured. GVB++ was incubated with NHS as a negative control. In all cases, the data were collected from three representative triplicate experiments ($n = 9$). Error bars denote standard deviation. *, ** and *** denote statistically significant differences, $p < 0.05$, 0.01 and 0.001, respectively, compared with IICs prepared in the absence of HRG or C1q, using Student's t test.

with the formation of IICs that were able to activate complement.

For IICs pre-formed in the presence of increasing HRG concentration, then incubated with NHS, a statistically significant dose-dependent increase in TCC concentration was observed (Fig. 7B). Thus, it appeared that IICs pre-formed in the presence of HRG were more effective activators of complement than those formed in the absence of HRG. In contrast, IICs pre-formed in the presence of 30 and 60 μ g/mL C1q showed statistically significant lower quantities of TCC (Fig. 7B). This would suggest that IICs formed as a consequence of incubation with C1q had a diminished ability to activate complement. For IICs pre-formed with both HRG and C1q, similar AU/mL TCC to IICs pre-formed with only C1q were obtained (Fig. 7B). Thus, the inhibitory effect seen when IICs were pre-formed with C1q appears to dominate over the enhancing effect seen for HRG. This also appears true when IICs are pre-formed at physiological concentrations of HRG (≈ 125 μ g/mL) and C1q (≈ 100 μ g/mL, Fig. 7B). To confirm that the above effects were specific for HRG and C1q, IICs prepared in the presence of 125 μ g/mL BSA were incubated with NHS and the quantity of TCC measured. Values for AU/mL very similar to those for TCCs prepared without effectors were obtained (Fig. 7B).

4. Discussion

The present study gives a deeper insight into the mechanisms by which HRG affects complement and immune function. Interactions of HRG with several components of the immune system have been reported previously. However, detailed information describing the functional consequences of many of these interactions is lacking.

We show here that HRG is present at appreciable concentrations in relevant patient materials, the plasma and synovial fluid of rheumatoid arthritis patients. Therefore, HRG might exert modulatory effects over serum immune function in an autoimmune disease setting. It is thought that HRG is exclusively synthesised in the liver (Koide et al., 1986), although it has also been found in megakaryocytes (Leung et al., 1983). Nonetheless, the presence of HRG in human tissues such as synovial fluid may reflect the acquisition of HRG from plasma (Hulett and Parish, 2000).

We show that HRG binds strongly and selectively to several complement proteins in buffers not supplemented with Zn^{2+} . Earlier reports (Gorgani et al., 1997) demonstrated that the binding affinities of HRG for IgG and C1q are affected by Zn^{2+} . However, the concentration of free Zn^{2+} *in vivo* is usually very low and tightly regulated (Tubek, 2007), suggesting that under normal conditions, the effect of this cation on HRG binding interactions should not be significant. Additionally, Zn^{2+} strongly inhibits complement (Blom et al., 2003b). This would make it impossible to reconcile binding data collected in the presence of Zn^{2+} with complement function data.

Having established that HRG binds to complement proteins *in vitro*, it was important to determine whether these interactions also occur *in vivo*. In both NHS and synovial fluid samples, HRG co-immunoprecipitates with C1q, FH and C4BP. These are the complement proteins to which HRG binds most strongly *in vitro*. Furthermore, we demonstrate here that HRG from serum bound to immobilised C1q, FH, C4BP and IgG. We conclude that *in vivo* at least some HRG is found complexed to the complement proteins C1q, FH and C4BP. It is unlikely that HRG was indiscriminately immunoprecipitated as a consequence of the formation of a ternary complex of HRG/C1q/IgG. We observed minimal immunoprecipitation of HRG when the irrelevant antibody to KCP was used.

Despite the fact that HRG interacts with several complement components it does not appear to affect complement function. Proteins interacting with C1q usually either activate C1q (if they bind to head domains) or inhibit C1q (if they bind to tails) but HRG showed neither effect. Binding to other complement components was also without effect as shown by both ELISA and haemolytic assay. We have noted discrepancies between the present results and those obtained in an earlier study (Chang et al., 1992). In our hands, HRG causes no change in serum-induced erythrocyte haemolysis rates even when pre-incubated with the serum, or when C1q pre-incubated with HRG was used to replete C1q-deficient serum. Earlier, it was suggested that the binding of HRG to C8 inhibits haemolysis and that HRG binds C9 and inhibits polymerization (Chang et al., 1992). However, none of this was reflected in our results.

Here, we confirm that both HRG and C1q bind to necrotic Jurkat T-cells, which are known to increase phagocytosis. Our present data extend these findings by showing that there is only a weak competitive effect between HRG and C1q for binding to necrotic cells. Nonetheless, it seems more plausible that the majority of C1q and HRG interact with different cell surface ligands. C1q and HRG may therefore act in synergy, mediating an accelerated clearance of necrotic material. HRG, when pre-incubated with necrotic cells, also enhanced complement activation and the subsequent deposition of C3b onto the cells. This enhancement could be explained in terms of the binding of serum C1q to cell-associated HRG, initiating a targeted or localised activation of complement. The effect

was only observed at serum concentrations 1% or less, suggesting that the importance of HRG in facilitating the localisation of C1q to necrotic cells becomes less as the concentration of C1q is increased.

Our data indicate that the growth of IICs of α -ovalbumin/ovalbumin is inhibited by HRG, but enhanced by C1q, corroborating earlier findings (Gorgani et al., 1997). The outward-facing B chains of the globular C1q heads bind to the interface between the F_c and F_{ab} regions on IgG molecules (Gaboriaud et al., 2004). Thus, C1q could facilitate the accelerated formation of larger IICs by tethering, in a polyvalent manner, ovalbumin/ α -ovalbumin units via these F_c/F_{ab} binding sites. HRG, on the other hand, binds via its N-terminal domain to a site in the F_{ab} region of IgG molecules (Gorgani et al., 1997). HRG is approximately half the molecular weight of IgG. Therefore, we propose that HRG can inhibit IIC growth by binding to IgG molecules in a monovalent manner, which would introduce steric hindrance effects. This, in turn, would perturb the Fc–Fc interactions that are important for IIC growth (Moller and Bak, 1984). Through steric hindrance effects HRG may also inhibit the binding of multiple IgG molecules to a single antigen molecule, again affecting the rate of IIC growth and final size attainable. When IICs are formed in the presence of both HRG and C1q, the enhancing effect of C1q appears to dominate. Because HRG binds to the tail region of C1q, the globular heads would remain unobstructed and could, hypothetically, still participate in F_c–F_c tethering, thus facilitating the observed enhancement in IIC growth.

Interestingly IICs pre-formed in the presence of C1q show a significantly diminished ability to activate complement. This seems plausible, as the C1q molecules associated with the antigen/antibody matrix would lower the number of free sites on the IgG molecules available to bind soluble C1q, as C1, from the serum. The net effect of this would be to lower the complement-activating potential of the IIC–C1q species.

Most of the former studies on the interaction of ICs with complement have focused on elucidating the mechanism by which serum complement can inhibit IC growth and facilitate IC dissolution. The consensus is that the C1 complex and the early classical pathway act to prevent the formation of very large and IICs (Schifferli et al., 1985) and that C3 cleavage products act to solubilise ICs through their covalent incorporation into the antigen/antibody matrix (Hong et al., 1984; Takata et al., 1984). Our findings, in contrast, describe another scenario, where C1q is directly incorporated into rapidly formed IICs. This incorporation leads to impaired complement activation capacity. Such a scenario has important implications in situations where C1q expression is upregulated, for example in nervous tissue in the latter stages of life and in response to neurodegenerative disorders (Pisalyaput and Tenner, 2008).

We report here that IICs pre-formed in the presence of HRG show a significantly elevated complement-activating potential. Because both HRG and IgG bind C1q, the affinity of soluble C1q for HRG-containing IICs may be greater than that of HRG-free IICs. In this way, the presence of HRG in IICs could facilitate complement activation via enhanced C1q binding. Alternatively, the presence of HRG in or on IICs may result in a change in supramolecular structure, which could hold the relevant binding site on IgG molecules in the antigen/antibody matrix in a more accessible conformation to bind soluble C1q.

For IICs that were pre-formed in the presence of both HRG and C1q, we found that complement activation is attenuated. Thus, the suggested inhibitory effect that is seen for IICs formed in the presence of C1q appears to dominate over any enhancement that is seen for IICs formed with HRG. It remains to be shown whether the incorporation of other serum proteins into immune complexes affects how they activate (and are cleared by) complement.

An earlier study reported that larger soluble immune complexes (SICs) are more efficient activators of the early components of the complement pathway (Doekes et al., 1984) than smaller SICs. How-

ever, this does not contradict our results, which show that the smaller IIC–HRGs are better complement activators than the larger IIC–C1qs. An examination of Doekes et al. (1984) indicates that, once SICs reach a size of approximately 4 antigen molecules (and the appropriate number of bound antibody molecules, which is dictated by the molecular weight of the antigen), a further increase in size results in only small increases in complement-activating potential. Because the IIC–HRGs examined in the present study scatter light, they had attained a size significantly larger than those examined in Doekes et al. (1984). Thus, the average size of the IIC–HRGs described here exceeds the size below which diminished complement activation can occur. Therefore, all of the IICs we examine here were large enough that the molecular size effect reported by Doekes et al. (1984) was unlikely to have affected the observed extents of complement activation.

Taken together, it appears that one of the major roles of HRG is to assist in the maintenance of normal immune function. This occurs through: (a) mediating the clearance of necrotic material, (b) preventing ICs from growing too large and insoluble, where they would otherwise become cytotoxic, and (c) increasing the complement-activating potential of ICs, leading to accelerated clearance by serum complement. In the context of autoimmune disorders, we therefore suggest that, over all, HRG acts to counter the effects of immune system dysregulation.

Future studies on HRG could be aimed at developing a therapeutic for autoimmune disorders that could enhance clearance rates of necrotic material, inhibit the formation of IICs and accelerate their complement-mediated removal. Another potentially attractive attribute to HRG is that it does not directly affect complement function, meaning that an HRG-derived therapeutic could be given in conjunction with a second complement-specific therapeutic.

Acknowledgements

The authors would also like to acknowledge the financial support of the Söderberg Foundation, the Swedish Research Council, the Swedish Foundation for Strategic Research and the Foundations of Österlund, King Gustav V's 80th Anniversary, Greta and Johan Kock, Knut and Alice Wallenberg and Inga-Britt and Arne Lundberg as well as research grants from the University Hospital in Malmö. Dr Manderson and Dr Martin are recipients of postdoctoral stipends from the Anna-Greta Crafoord foundation. The authors thank Professor CR Parish (Division of Immunology and Genetics, John Curtin School of Medical Research, Australian National University, Canberra, ACT 2601, Australia) for providing the α HRG mAb. We also thank Frida Bergström, who purified several of the complement proteins used in this study.

References

- Arnett, F.C., Edworthy, S.M., Bloch, D.A., McShane, D.J., Fries, J.F., Cooper, N.S., Healey, L.A., Kaplan, S.R., Liang, M.H., Luthra, H.S., et al., 1988. The American Rheumatism Association 1987 revised criteria for the classification of rheumatoid arthritis. *Arthritis Rheum.* 31, 315–324.
- Blom, A.M., Kask, L., Dahlback, B., 2003a. CCP1–4 of the C4b-binding protein alpha-chain are required for factor I mediated cleavage of complement factor C3b. *Mol. Immunol.* 39, 547–556.
- Blom, A.M., Kask, L., Ramesh, B., Hillarp, A., 2003b. Effects of zinc on factor I cofactor activity of C4b-binding protein and factor H. *Arch. Biochem. Biophys.* 418, 108–118.
- Blom, A.M., Villoutreix, B.O., Dahlback, B., 2003c. Mutations in alpha-chain of C4BP that selectively affect its factor I cofactor function. *J. Biol. Chem.* 278, 43437–43442.
- Chang, N.S., Leu, R.W., Rummage, J.A., Anderson, J.K., Mole, J.E., 1992. Regulation of complement functional efficiency by histidine-rich glycoprotein. *Blood* 79, 2973–2980.
- Charlesworth, J.A., Endre, Z.H., Pussell, B.A., Yasmeen, D., Peake, P.W., 1982. Complement behavior in infectious mononucleosis: possible mechanisms for the prevention of immune complex injury. *J. Infect. Dis.* 145, 505–513.

- Ciurana, C.L., Zwart, B., van Mierlo, G., Hack, C.E., 2004. Complement activation by necrotic cells in normal plasma environment compares to that by late apoptotic cells and involved predominantly IgM. *Eur. J. Immunol.* 34, 2609–2619.
- Crossley, L.G., Porter, R.R., 1980. Purification of the human complement control protein C3b inactivator. *Biochem. J.* 191, 173–182.
- Dahlbäck, B., 1983. Purification of human C4b-binding protein and formation of its complex with vitamin K-dependent protein S. *Biochem. J.* 209, 847–856.
- Doekes, G., van Es, L.A., Daha, M.R., 1984. Binding and activation of the first complement component by soluble immune complexes: effect of complex size and composition. *Scand. J. Immunol.* 19, 99–110.
- Easterbrook-Smith, S.B., Vandenberg, R.J., Alden, J.R., 1988. The role of Fc:Fc interactions in insoluble immune complex formation and complement activation. *Mol. Immunol.* 25, 1331–1337.
- Elward, K., Griffiths, M., Mizuno, M., Harris, C.L., Neal, J.W., Morgan, B.P., Gasque, P., 2005. CD46 plays a key role in tailoring innate immune recognition of apoptotic and necrotic cells. *J. Biol. Chem.* 280, 36342–36354.
- Gaboriaud, C., Thielens, N.M., Gregory, L.A., Rossi, V., Fontecilla-Camps, J.C., Arlaud, G.J., 2004. Structure and activation of the C1 complex of complement: unraveling the puzzle. *Trends Immunol.* 25, 368–373.
- Gorgani, N.N., Parish, C.R., Altin, J.G., 1999. Differential binding of histidine-rich glycoprotein (HRG) to human IgG subclasses and IgG molecules containing kappa and lambda light chains. *J. Biol. Chem.* 274, 29633–29640.
- Gorgani, N.N., Parish, C.R., Easterbrook Smith, S.B., Altin, J.G., 1997. Histidine-rich glycoprotein binds to human IgG and C1q and inhibits the formation of insoluble immune complexes. *Biochemistry* 36, 6653–6662.
- Gorgani, N.N., Smith, B.A., Kono, D.H., Theofilopoulos, A.N., 2002. Histidine-rich glycoprotein binds to DNA and Fc gamma RI and potentiates the ingestion of apoptotic cells by macrophages. *J. Immunol.* 169, 4745–4751.
- Haeblerli, A., 1992. Human Protein Data. VCH, Weinheim, New York, Basel, Cambridge.
- Heinegård, D., Larsson, T., Sommarin, Y., Franzen, A., Paulsson, M., Hedbom, E., 1986. Two novel matrix proteins isolated from articular cartilage show wide distributions among connective tissues. *J. Biol. Chem.* 261, 13866–13872.
- Hong, K., Takata, Y., Sayama, K., Kozono, H., Takeda, J., Nakano, Y., Kinoshita, T., Inoue, K., 1984. Inhibition of immune precipitation by complement. *J. Immunol.* 133, 1464–1470.
- Hulett, M.D., Parish, C.R., 2000. Murine histidine-rich glycoprotein: cloning, characterization and cellular origin. *Immunol. Cell Biol.* 78, 280–287.
- Jancar, S., Sanchez Crespo, M., 2005. Immune complex-mediated tissue injury: a multistep paradigm. *Trends Immunol.* 26, 48–55.
- Jones, A.L., Hulett, M.D., Parish, C.R., 2005a. Histidine-rich glycoprotein: a novel adaptor protein in plasma that modulates the immune, vascular and coagulation systems. *Immunol. Cell Biol.* 83, 106–118.
- Jones, A.L., Poon, I.K., Hulett, M.D., Parish, C.R., 2005b. Histidine-rich glycoprotein specifically binds to necrotic cells via its amino-terminal domain and facilitates necrotic cell phagocytosis. *J. Biol. Chem.* 280, 35733–35741.
- Kohl, J., 2001. Anaphylatoxins and infectious and non-infectious inflammatory diseases. *Mol. Immunol.* 38, 175–187.
- Koide, T., Foster, D., Yoshitake, S., Davie, E.W., 1986. Amino acid sequence of human histidine-rich glycoprotein derived from the nucleotide sequence of its cDNA. *Biochemistry* 25, 2220–2225.
- Leung, L.L., Harpel, P.C., Nachman, R.L., Rabellino, E.M., 1983. Histidine-rich glycoprotein is present in human platelets and is released following thrombin stimulation. *Blood* 62, 1016–1021.
- Moller, N.P., Bak, J.F., 1984. A general theory for the precipitin reaction—based on Fc-mediated precipitation. *Acta Pathol. Microbiol. Immunol. Scand. [C]* 92, 237–246.
- Mollnes, T.E., Lea, T., Froland, S.S., Harboe, M., 1985. Quantification of the terminal complement complex in human plasma by an enzyme-linked immunosorbent assay based on monoclonal antibodies against a neoantigen of the complex. *Scand. J. Immunol.* 22, 197–202.
- Paques, E.P., Huber, R., Priess, H., Wright, J.K., 1979. Isolation of the globular region of the subcomponent q of the C1 component of complement. *Hoppe Seylers Z. Physiol. Chem.* 360, 177–183.
- Pisalyaput, K., Tenner, A.J., 2008. Complement component C1q inhibits beta-amyloid- and serum amyloid P-induced neurotoxicity via caspase- and calpain-independent mechanisms. *J. Neurochem.* 104, 696–707.
- Reid, K.B., Porter, R.R., 1976. Subunit composition and structure of subcomponent C1q of the first component of human complement. *Biochem. J.* 155, 19–23.
- Schifferli, J.A., Steiger, G., Schapira, M., 1985. The role of C1, C1-inactivator and C4 in modulating immune precipitation. *Clin. Exp. Immunol.* 60, 605–612.
- Sjöberg, A., Önerfjord, P., Mörgelin, M., Heinegård, D., Blom, A.M., 2005. The extracellular matrix and inflammation: fibromodulin activates the classical pathway of complement by directly binding C1q. *J. Biol. Chem.* 280, 32301–32308.
- Smith, B.O., Mallin, R.L., Krych-Goldberg, M., Wang, X., Hauhart, R.E., Bromek, K., Uhrin, D., Atkinson, J.P., Barlow, P.N., 2002. Structure of the C3b binding site of CR1 (CD35), the immune adherence receptor. *Cell* 108, 769–780.
- Takata, Y., Tamura, N., Fujita, T., 1984. Interaction of C3 with antigen-antibody complexes in the process of solubilization of immune precipitates. *J. Immunol.* 132, 2531–2537.
- Tenner, A.J., Lesavre, P.H., Cooper, N.R., 1981. Purification and radiolabeling of human C1q. *J. Immunol.* 127, 648–653.
- Tubek, S., 2007. Zinc supplementation or regulation of its homeostasis: advantages and threats. *Biol. Trace Elem. Res.* 119, 1–9.